

Supplementary Information

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Supplementary Information

Manuscript: Interpretable Parameter-Space Mapping of Multiplexed Coding in a Two-Compartment Burst-Propagation Model

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Supplementary Methods

S1. Default BPM parameters

The default parameters used in the burst-propagation model were: $E_L = -70$ mV, $V_T = -50$ mV, $V_r = -70$ mV, $\tau_s = 16$ ms, $\tau_d = 7$ ms, $C_s = 370$ pF, $C_d = 170$ pF, $\tau_{ws} = 100$ ms, $\tau_{wd} = 30$ ms, $b = 200$ pA, $a = 13$ nS, $E_d = -38$ mV, and $D_d = 6$ mV.

S2. Input protocol

The somatic and dendritic compartments received square-wave inputs with a period of 200 ms. The somatic input used a low-state ratio of 0.5, and the dendritic input used a low-state ratio of 0.4. The TreeNet analysis reported in the main text used a phase-excluded dataset.

S3. Noise implementation

Independent Ornstein–Uhlenbeck noise processes were added to the somatic and dendritic inputs. The noise relaxation parameter was fixed at $\alpha = 1$, and the noise amplitude was controlled by σ .

S4. Event, burst, and multiplexing definitions

Spikes were detected from somatic threshold crossings. Events and bursts were identified using the interspike-interval rule described in the main text. Popula-

tion output was binned at 1-ms resolution. Burst fraction was defined as

$$\text{BF}(t) = \begin{cases} \frac{\text{BR}(t)}{\text{ER}(t)}, & \text{ER}(t) > 0, \\ 0, & \text{ER}(t) = 0. \end{cases}$$

The multiplexing metric was defined as

$$M = \frac{\text{Corr}(\text{ER}(t), I_s(t)) + \text{Corr}(\text{BF}(t), I_d(t))}{2}.$$

S5. TreeNet dataset

Supplementary Data 1 contains the phase-excluded TreeNet dataset. The original compiled dataset contained 1000 simulated configurations. Rows with undefined correlation-based output metrics were excluded before TreeNet fitting, leaving 990 complete observations.

S6. TreeNet settings

TreeNet regression used squared-error loss, 5-fold cross-validation, a learning rate of 0.1, a subsample fraction of 0.7, a maximum of 6 terminal nodes per tree, a minimum terminal node size of 3, and five predictors: σ , $I_{d,0}$, $I_{s,0}$, g_s , and c_d . A total of 300 trees were grown, and the optimal ensemble size was 258 trees.